Do Now · retrieval

Lesson 147 · Firewalls, Proxies and Encryption: Layered Defence

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What attack floods a service with requests to reduce availability?

Q2

What is a foreign key used for?

Q3

Which storage type is volatile?

Q4

Which layer in TCP/IP handles IP routing?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Filtering boundary — Firewall applies rules to traffic entering or leaving a network.

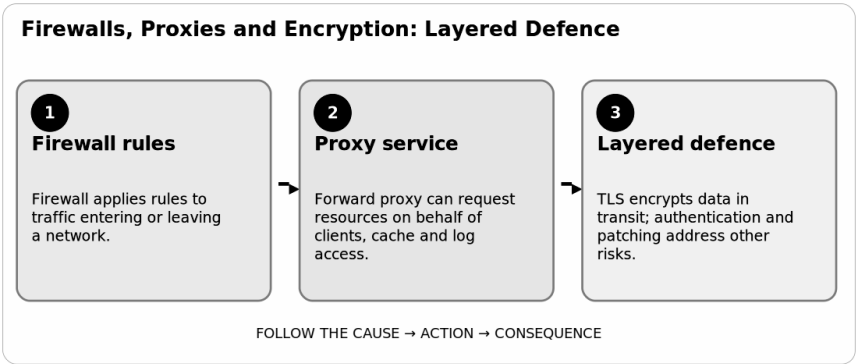
Core idea 2

Mediating requests — Forward proxy can request resources on behalf of clients, cache and log access.

Core idea 3

Protecting content — TLS encrypts data in transit; authentication and patching address other risks.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

firewall — Rule-based network traffic access control.

proxy server — Server that acts as intermediary for client requests.

TLS — Protocol protecting transport of application data.

defence in depth — Overlapping safeguards against different failures.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A school wants to limit outgoing web browsing while protecting pupils' login details over public Wi-Fi. Select appropriate controls.

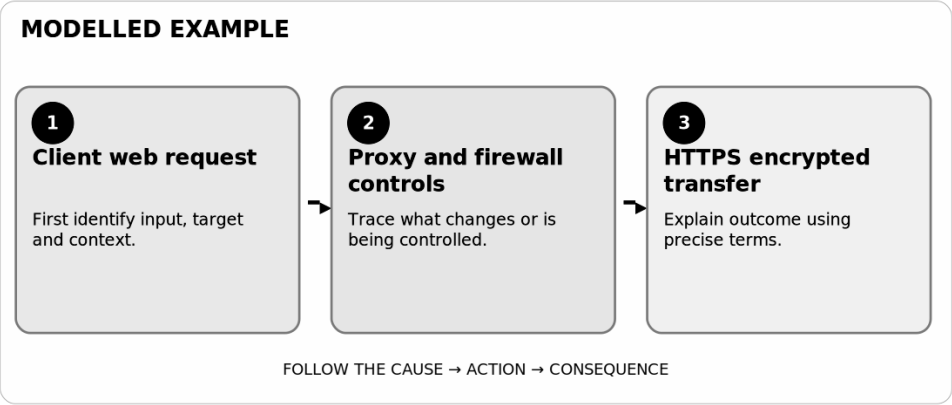
Follow the worked method

1. Client web request — Identify the facts and make an initial prediction.

2. Proxy and firewall controls — Explain the mechanism or trace the intermediate stage.

3. HTTPS encrypted transfer — A managed web proxy can filter/log permitted requests; a firewall can restrict traffic flows; HTTPS/TLS encrypts the connection so local eavesdroppers cannot read credentials. Policy, MFA and endpoint protection address further threats.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Traffic allowed on port 443
Identify the facts and make an initial prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State functions of firewall, forward proxy and TLS, one network example each. [6]

Question 2

Explain why encrypted traffic is not the same as authenticated trustworthy content. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Recommend a layered defence for a sixth-form network with remote learners, justifying four different controls. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A forward proxy and a firewall necessarily perform identical tasks.

Use key vocabulary from memory

1. Explain firewall in your own words.

2. Explain proxy server in your own words.

3. Explain TLS in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.